

# ED + MCMC Direction Reconstruction

## 3 Categories, 120 Events

### PERFORMANCE SUMMARY

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Cosine of Angle (True vs Reconstructed Direction)

Mean: 0.0519  
Median (p50): 0.1139  
68th percentile: 0.4863 ← PRIMARY METRIC  
90th percentile: 0.7995  
Std deviation: 0.5734

Angular Error (degrees)

Mean: 86.36°  
Median: 83.46°  
68th percentile: 109.68°  
90th percentile: 134.47°

### INTERPRETATION

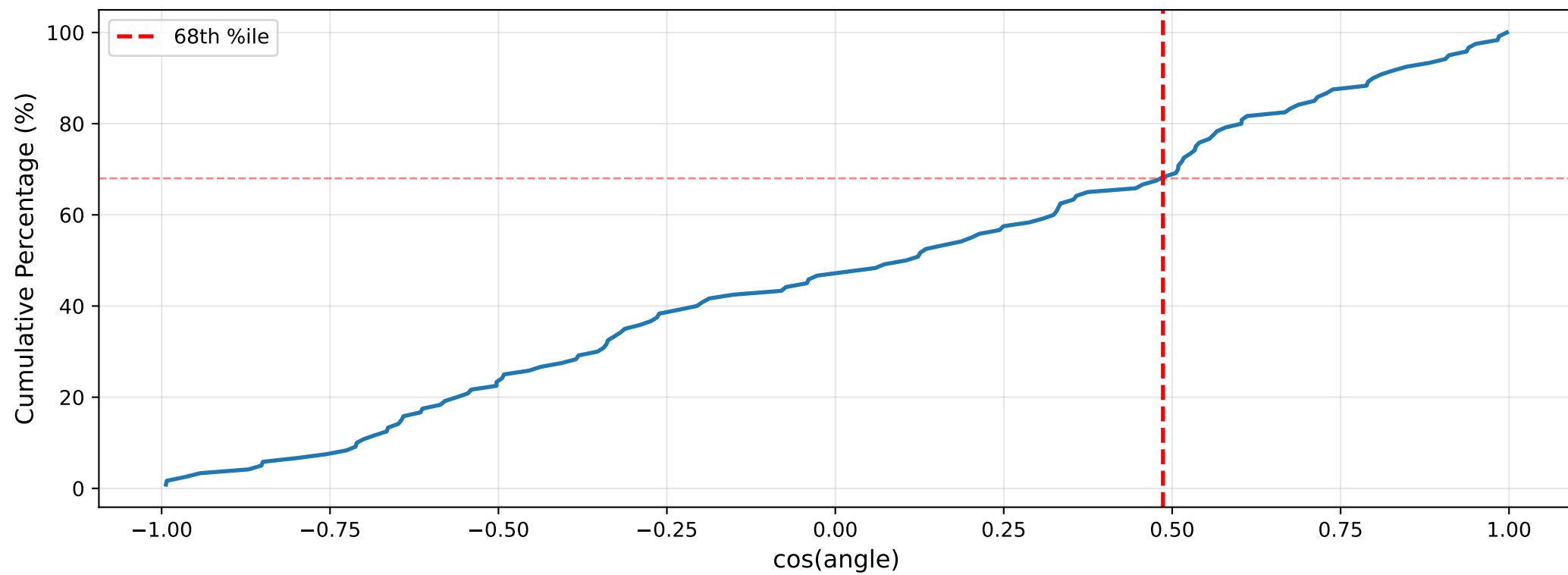
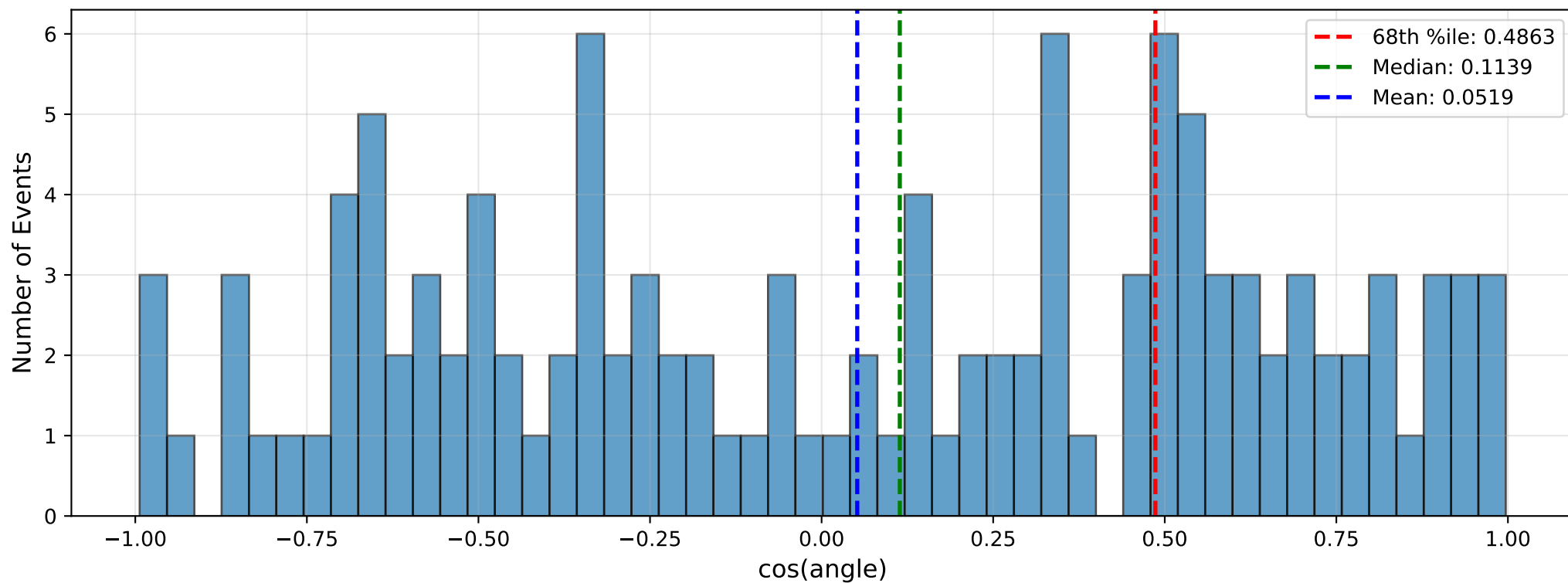
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68% of events have  $\cos(\text{angle}) \geq 0.4863$   
→ 68% of events within 60.90° angular resolution

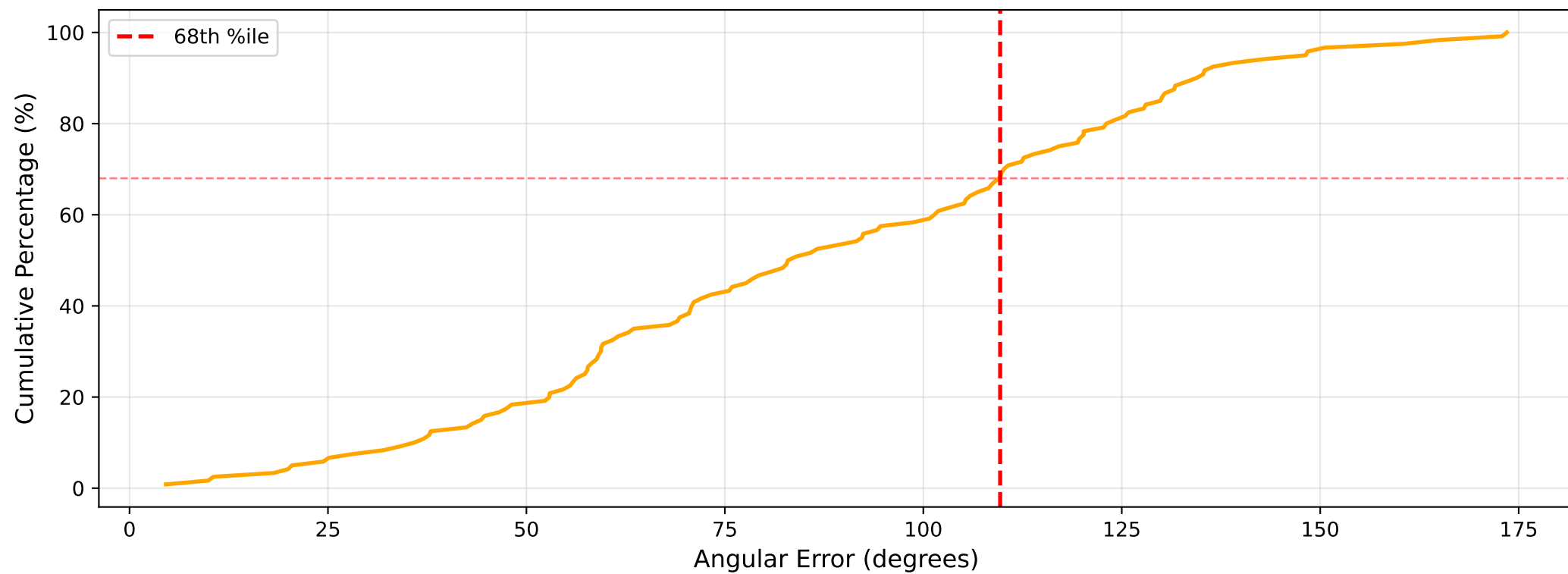
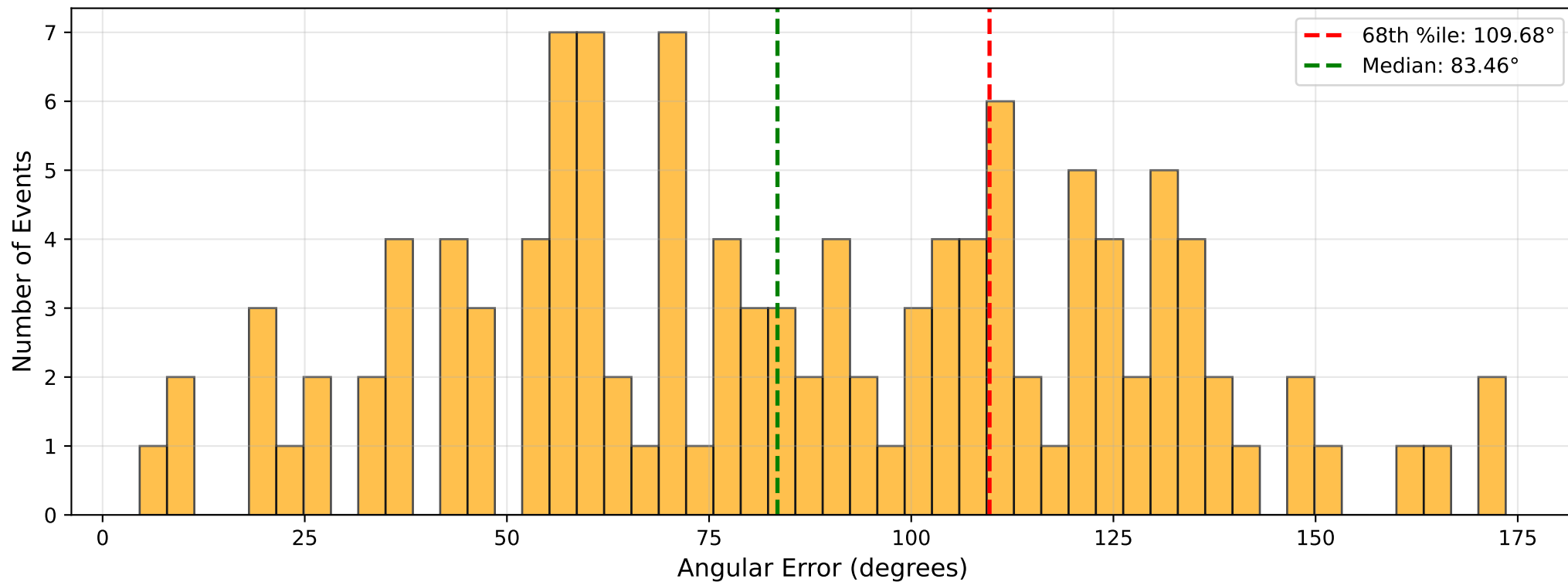
Perfect pointing:  $\cos(\text{angle}) = 1.0$  (0°)  
Random pointing:  $\cos(\text{angle}) \sim 0.0$  (90°)

Dataset: 3 categories × ~40 events/cat

**Distribution of  $\cos(\text{angle})$  Between True and Reconstructed**



# Angular Error Distribution (degrees)



Performance by Category

